Lap Time Prediction in Formula 1

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December 2024

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Abstract

Predicting lap times in Formula 1 racing presents a complex challenge due to the multifaceted nature of the sport. This project develops a predictive model using the 2023 Formula 1 season's comprehensive dataset. By integrating advanced data collection methods through the FastF1 library, the study captures critical variables including track geometry, weather conditions, driver metrics, and race telemetry.

The research employed three machine learning models—Linear Regression, Decision Tree Regressor, and Random Forest Regressor—to predict lap times. The Decision Tree Regressor demonstrated superior performance, achieving a Mean Absolute Error of 0.9487 and an R-squared score of 0.9478.

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1 Data Collection

This Python project utilizes the FastF1 library to collect and analyze comprehensive Formula 1 racing data for the 2023 season.

1.1 Key Modules and Their Functionalities

These modules are stored in the `helper` folder.

1. `RaceCalendar` Module: Retrieves the official F1 race schedule for a specific year (excluding the pre-season testing). Extracts key event details including:
    - Round number
    - Event name
    - Location
    - Country
    - Event date
2. `TrackGeometryAnalyzer` Module: Performs advanced track geometric analysis by:
    (a) Calculating track length
    (b) Analyzing elevation profiles
        - Maximum elevation
        - Minimum elevation
        - Total elevation change
        - Elevation standard deviation
    (c) Assessing track curvature:
        - Number of corners
        - Total curvature
        - Maximum curvature
        - Curvature standard deviation
3. `TrackWeatherAnalyzer` Module: Captures time-series weather data and engineers sophisticated weather-related features
    - Air temperature
    - Track temperature
    - Rainfall
    - Humidity-wind interaction
    - Wind chill factor
    - Surface grip index
4. `DriverMetrics` Module: Retrieves driver points before the current event as a metric of their performance over the entire season as well as qualifying performance metrics to measure performance on this track.
    - Best lap time (Entire Time and Delta from Fastest Lap in Qualifying session)
    - Qualifying position
    - Sector-wise lap times

- Tire compound used
- Points before the start of the weekend

5. `RaceTelemetry` Module: Captures lap-by-lap telemetry data for each driver.
    - Lap start times
    - Lap times
    - Tire information (Compound and Tyre Age)
    - Sector times
    - Track status
    - Personal best lap indicators

1.2 Pipeline

- `store_race_calendar()`:
  Retrieves and saves the race calendar for the specified year(s), with a `year` column to the data
- `store_track_geometry()`:
  For each race in the race calendar, uses `TrackGeometryAnalyzer` to calculate track characteristics and combines race information with track geometry data.
- `store_track_weather()`:
  For each race in the race calendar, uses `TrackWeatherAnalyzer` to calculate weather characteristics and combines race information with weather data.
- `store_driver_metrics()`:
  For each race in the the race calendar, retrieves driver performance metrics. Calculates `BestLapTimeDelta` (difference from fastest lap in the respective qualifying session).
- `store_race_telemetry()`: For each race in the the race calendar,collects race telemetry data, dropping rows with missing data or laps where there was an incident on track.

2 Data Pre-processing

- For all databases, we drop columns that will not be used in the analysis.
- For time series data, we quantize the time to the nearest minute in the datasets using `quantize_time()` so that we can join on the time later.

3 Data Processing

We now combines multiple Formula 1 datasets into a comprehensive, machine learning-ready dataset.

1. Join data from multiple sources
    - Track Weather
    - Track Geometry
    - Race Telemetry

- Driver Metrics

2. Calculates derived features like `LapsLeft`
   We use `LapsLeft` as a metric for the amount of fuel left in the car as this affects the weight of the car, which has tremendous effect of the speeds and hence the lap times.

3. Categorical Encoding using `OneHotEncoder` for tire compound data, multiplied tire life to account for the interaction between these variables

The final output includes features such as:

- Basic identifiers (year, round, driver number)
- Tire information (compound, life)
- Weather conditions
- Track geometry metrics
- Driver performance indicators
- Lap time as target variable

4 Model Comparison

4.1 Evaluation Context

- Loss Function: Mean Absolute Error (MAE):
  Measures average absolute difference between predicted and actual values.
  We use this because it is less sensitive to outliers compared to MSE.
- Metrics: R^2
  Other than MAE, we use R^2 to measure the share of the data explained by the model.
- Data Preparation:
  Remove identifier columns like `year`, `round`, and `DriverNumber`.
  Split data into training and testing sets (80/20 split).

4.2 Models Considered

We consider the following models for prediction:

1. Linear Regression:
   Pros:
   - Provides a good baseline

   Cons:
   - Assumes linear relationships in fundamentally non-linear domain
   - Cannot capture complex performance interactions

2. Decision Tree
   Pros:
   - Very explainable

Cons:

- Performance may degrade across different tracks/conditions

3. Random Forest
   Pros:

   - Captures non-linear interactions
   - Reduces individual model biases

   Cons:

   - Large training time

4.3 Performance Evaluation

4.3.1 Linear Regression

- Optimization: Ordinary Least Squares
- Mean Absolute Error: 3.9319
- R-squared Score: 0.7397

4.3.2 Decision Tree Regressor

- Splitting Criterion: Mean Absolute Error
- Hyper-parameters Tuned:
  - `max_depth` = 20: Maximum tree depth
  - `min_samples_split` = 2: Minimum samples to create a split
  - `min_samples_leaf` = 10: Minimum samples in a leaf node
- Mean Absolute Error: 0.9487
- R-squared Score: 0.9478

4.3.3 Random Forrest Regressor

- Splitting Criterion: Mean Absolute Error
- Hyper-parameters Tuned:
  - `n_estimators` = 100: Number of trees
  - `max_depth` = 15: Maximum tree depth
  - `min_samples_split` = 20: Minimum samples to create a split
  - `min_samples_leaf` = 100: Minimum samples in a leaf node
- Mean Absolute Error: 1.6796
- R-squared Score: 0.8922

As we can clearly see, the most important feature in both cases is the length of the track, followed by variation in elevation, the curvature of the track, i.e., the turns and the number of turns.

5 Challenges in Prediction

Formula 1 is a domain where traditional predictive modeling encounters numerous unique challenges:

5.1 Technological Variability

- Constant Technological Evolution: F1 regulations change annually
- Car Development Mid-Season: Teams continuously upgrade vehicle components

5.2 Driver-Specific Factors

- Individual Driving Styles & Tire Management Strategies: Each driver has unique approach to track management to deal with the complex interaction between compound selection and track conditions
- Psychological Factors: Mental state, confidence, and race strategy impact performance
- Physical Conditioning: Driver fitness plays a critical role in consistent lap times

5.3 Unpredictable External Conditions

- Weather Dynamics: Rapid and localized weather changes
- Track Temperature Variations: Surface conditions change dramatically during race weekends
- Tire Performance Degradation: Non-linear tire wear characteristics
- Unique Circuit Characteristics: Monaco's tight corners vs. Monza's high-speed straights

6 Conclusion

The Decision Tree Regressor emerged as the most accurate model, achieving a Mean Absolute Error of 0.9487 seconds and R^2 Score of 0.9478. While we have achieved a very high R^2, the MAE of 0.9487 seconds is very significant in a sport where the difference of some milliseeconds can cost millions of dollars.

While machine learning can provide valuable insights into lap time prediction, the inherent complexity of Formula 1 racing means that models must continuously adapt to technological, human, and environmental variables. The study demonstrates both the potential and limitations of predictive modeling in this high-stakes, dynamic sport.